

**MATH 6908A- READING COURSE:
GEOMETRIC REPRESENTATION THEORY
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QINGYUAN JIANG

Course Description. This graduate reading course studies geometric representation theory through selected foundational papers. We begin with Hall algebras over finite fields and homological convolution, then study Nakajima’s constructions of Heisenberg and Kac–Moody representations using Hilbert schemes of points and quiver varieties, respectively. We conclude with Kleinian singularities, the derived McKay correspondence, and Kapranov–Vasserot’s Euler-characteristic Hall algebra.

Meeting Time and Venue. Tuesdays 4:30–7:20 pm, Room 5564 (Lift 27–28).

Pre-requisite. Algebraic geometry (MATH 5251, MATH 5261) and basic homological algebra, including schemes, coherent sheaves, and Ext groups. Familiarity with derived categories and GIT is useful. Prior knowledge of Hall algebras or quiver varieties is not assumed.

Course Outline. Presentations should emphasize main constructions, precise theorems, and worked examples, distinguishing results proved from those quoted.

Part I: Hall Algebras and Geometric Correspondences.

- **Topic 1: Hall algebras over finite fields.** Review quiver representations and extensions over $\mathbb{F}_q$. Define Hall multiplication by counting subobjects in a finitary abelian category and prove associativity using flags. Compute examples for $1 \rightarrow 2$ and the nilpotent Jordan quiver; briefly state relations with quantum group.

Reading: [Sch], §§1.1–1.3 and 2.1–2.2; supplementary: §§2.3–2.4 and introductory material in §3.

- **Topic 2: Homological convolution and Hilbert schemes.** Introduce Borel–Moore homology, fundamental classes, proper pushforward, and the intersection operations defining convolution. Explain correspondences as operators and the properness conditions on supports. Introduce Hilbert schemes of points, the Hilbert–Chow morphism, and incidence correspondences.

Reading: [CG], §§2.6–2.7; [Gin], §§6.1–6.4; [Nak-Hilb], §§2–3.

Part II: Nakajima’s Heisenberg and Kac–Moody Representations.

- **Topic 3: Nakajima’s Heisenberg action and Fock space.** Construct creation and annihilation operators on the homology of Hilbert schemes. State the graded Heisenberg relations, compare the two orders of composition, and explain the diagonal contribution and Fock-space interpretation. Treat technical intersection calculations selectively.

Reading: [Nak-Hilb], §§3–4. Supplementary: [Nak-Book], Chapter 8.

- **Topic 4: Quiver varieties and Kleinian examples.** Introduce doubled quivers, preprojective algebras, framing, moment maps, and stability. Explain $\mathfrak{M}(\mathbf{v}, \mathbf{w}) \rightarrow \mathfrak{M}_0(\mathbf{v}, \mathbf{w})$, using cotangent Grassmannians and Hilbert schemes of $\mathbb{A}^2$ as examples. For finite $G \subset \mathrm{SL}_2(\mathbb{C})$, briefly introduce $\mathbb{C}^2/G$, its minimal resolution, the exceptional ADE curves, and the affine McKay graph; illustrate the A_1 case by $T^*\mathbb{P}^1$. Defer derived McKay to Topic 6.

Reading: [Gin], selected portions of §§2–5, including §4.6; the geometric setup in [Nak-Quiver]. Optional ALE context: [Nak-ALE].

- **Topic 5: Hecke correspondences and Kac–Moody actions.** For quivers without edge loops, introduce Cartan matrices and Chevalley generators. Construct Hecke operators, explain the $\mathfrak{sl}_2$ relation, and state the $U(\mathfrak{g})$ -action and highest-weight construction on the top homology of the Lagrangian fibres over 0.

Reading: [Nak-Quiver], §§5–9 and §10 for the highest-weight construction; [Gin], §7.

Part III: Derived McKay and Kapranov–Vasserot’s Hall Algebras.

- **Topic 6: Derived McKay correspondence.** Return to the minimal resolution $Y \rightarrow \mathbb{C}^2/G$ from Topic 4. Explain the universal-family Fourier–Mukai functor, state the derived equivalence, and present selected proof ideas. Examine images of simple objects in the A_1 example, introducing hearts and shifts as needed.

Reading: [KV1], §§1–2. Supplementary stability background: selected portions of [Bri].

- **Topic 7: Kapranov–Vasserot’s Euler-characteristic Hall algebra.** Define Hall multiplication by integration of constructible functions against compactly supported Euler characteristic, contrasting it with Topic 1. For an ADE configuration $C = \bigcup_i C_i$ of (-2) -curves on a smooth complex surface S , explain the homomorphism

$$U(\mathfrak{g}_C^+) \longrightarrow H(S, C) \quad e_i \longmapsto [\mathcal{O}_{C_i}].$$

Here $\mathfrak{g}_C^+$ is the positive Lie subalgebra associated with the graph, $H(S, C)$ is the Euler-characteristic Hall algebra of sheaves on S supported on C , and $[\mathcal{O}_{C_i}]$ denotes a characteristic function. Explain Lusztig’s Serre-relation input and the role of compatible hearts under derived McKay. This is constructible-function, not cohomological, Hall theory.

Reading: [KV1], §§3–4.

Optional Topic: Hall Algebras of Curves. If time permits, study torsion sheaves, line bundles, elementary modifications, and Hall products on $\mathbb{P}^1$ over a finite field, with an introduction to the quantum-loop-algebra connection. *Reading:* [Sch], §§4.1–4.3; selected material from [Kap].

REFERENCES

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